



IEF SKAT (Industrial Equipment Factory SKAT)

DIGITAL DIELECTRIC TEST SYSTEMS

SKAT-70C / SKAT-100C Series

AC/DC High-Voltage Insulation Test Sets

TECHNICAL PASSPORT

Document No. STSK.411728.014-2022 PS | English translation, Rev. 1.0

Applies to all modifications: SKAT-70C, SKAT-70CA, SKAT-70C-K, SKAT-70CA-K, SKAT-70C-S, SKAT-70CA-S, SKAT-70C-H, SKAT-70CA-H, SKAT-100C, SKAT-100CA, SKAT-100C-S, SKAT-100CA-S

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1 General Product Information

The SKAT series digital dielectric test systems (hereafter, "the instrument") generate a 50 Hz sinusoidal AC high voltage and a positive- or negative-polarity DC high voltage, and measure the RMS values of AC voltage and current and the amplitude/average values of DC voltage and current, for high-voltage testing and insulation diagnostics of power cables, surge arresters, solid dielectrics, protective equipment and other objects and materials that require high-voltage testing.

The instrument is available in 12 modifications, differing in the design of the control unit (CU) and high-voltage unit (HVV), DC output polarity, maximum DC/AC voltage and current output/measurement values, and permissible measurement uncertainty limits.

- **SKAT-70C, SKAT-70CA, SKAT-100C, SKAT-100CA** — CU in a portable case, HVU in a steel enclosure.
- **SKAT-70C-K, SKAT-70CA-K** — CU in a portable case, HVU in a plastic enclosure.
- **SKAT-70C-H, SKAT-70CA-H** — CU in a metal enclosure with carrying handle, HVU in a steel enclosure.
- **SKAT-70C-S, SKAT-70CA-S, SKAT-100C-S, SKAT-100CA-S** — CU in a 19" rack-mount enclosure, HVU in a steel enclosure.

DC output polarity is negative for the SKAT-70C, SKAT-70C-K, SKAT-70C-S, SKAT-70C-H, SKAT-100C and SKAT-100C-S modifications, and positive for the SKAT-70CA, SKAT-70CA-K, SKAT-70CA-S, SKAT-70CA-H, SKAT-100CA and SKAT-100CA-S modifications.

Manufacturer: IEF SKAT (Industrial Equipment Factory SKAT)

Manufacturer's address: 37 Poddubnogo St., Volgograd 400040, Russian Federation

Type-approval certificate (Russian Federation) No. 87761-22, issued by the Federal Agency on Technical Regulating and Metrology (Rosstandart); valid through 20 December 2027.

The instrument complies with GOST 12.2.091-2012, GOST 22261-94, TR TS 004/2011, TR TS 020/2011, SanPiN 2.2.4.1191-03 and STSK.411728.014 TU.

2 Embedded Software

Table 1 — Software identification data

Identification data	Value
Software identifier	SKAT-C-series
Version number (software ID)	1.0 or later
Digital software identifier	—

The firmware is stored in the microcontroller's non-volatile memory and is metrologically significant; the instrument's metrological characteristics are rated with the influence of the firmware taken into account. The instrument design precludes unauthorized interference with the firmware or measurement data; the firmware can be verified, installed or reinstalled only at the manufacturer's facility using dedicated tools. Per R 50.2.077-2014, the firmware's protection level against unintentional and deliberate modification is **"high."**

3 Technical Specifications

Table 2 — Metrological characteristics

Characteristic	SKAT-70C family ¹	SKAT-100C family ²
AC output voltage range, 50 Hz sinusoidal, RMS, kV	1 to 51	1 to 71
DC output voltage range, either polarity, kV	1 to 71	1 to 101
AC voltage measurement range, RMS, kV	2 to 50	3 to 70
DC voltage measurement range, either polarity, amplitude, kV	3 to 70	5 to 100
Permissible relative measurement uncertainty, AC RMS voltage, %	$\pm(2.0+0.04 \cdot ((50/U)-1))^*$	$\pm(2.0+0.04 \cdot ((70/U)-1))^*$
Permissible relative measurement uncertainty, DC voltage (amplitude), %	$\pm(2.0+0.04 \cdot ((70/U)-1))^*$	$\pm(2.0+0.04 \cdot ((100/U)-1))^*$
Maximum AC output current, mA	50	35
Maximum DC output current, mA	20	15
AC current measurement range, RMS, mA	0.1 to 50	0.1 to 35
DC current measurement range, either polarity, average, mA	0.1 to 20	0.1 to 15
Permissible relative measurement uncertainty, AC current, %	$\pm(2.0+0.1 \cdot ((50/I)-1))^*$	$\pm(2.0+0.1 \cdot ((35/I)-1))^*$
Permissible relative measurement uncertainty, DC current, %	$\pm(2.0+0.1 \cdot ((20/I)-1))^*$	$\pm(2.0+0.1 \cdot ((15/I)-1))^*$
Test DC voltage ripple factor, %, max.	3	
AC test voltage sinusoidal distortion factor, %, max.	5	

1 SKAT-70C, SKAT-70CA, SKAT-70C-K, SKAT-70CA-K, SKAT-70C-S, SKAT-70CA-S, SKAT-70C-H, SKAT-70CA-H.

2 SKAT-100C, SKAT-100CA, SKAT-100C-S, SKAT-100CA-S. *U – measured voltage, kV; I – measured current, mA.

Table 3 — General technical characteristics

Characteristic	Value
Supply voltage, V	198 to 242
Supply frequency, Hz	49 to 51
Supply voltage distortion factor, %, max.	5
Maximum apparent power consumption, V·A	2800
Maximum AC output apparent power, V·A, not less than	2000
Maximum DC output power, W, not less than	700
Mean service life, years, not less than	10
Mean time between failures, h, not less than	8000
Mean time to restore, h, not less than	8

Table 3a — Overall dimensions and mass, max.

Unit / Modification	H×W×L, mm	Mass, kg
Control unit — SKAT-70C, 70CA, 70C-K, 70CA-K, 100C, 100CA	420×340×220	14
Control unit — SKAT-70C-S, 70CA-S, 100C-S, 100CA-S	180×450×300	17.5
Control unit — SKAT-70C-H, 70CA-H	200×420×350	15
HV unit — SKAT-70C, 70CA, 70C-S, 70CA-S, 70C-H, 70CA-H	650×375×360	29
HV unit — SKAT-70C-K, 70CA-K	660×460×350	35
HV unit — SKAT-100C, 100CA, 100C-S, 100CA-S	1020×420×650	80

Conditions of use: ambient temperature –10 to +40 °C; relative humidity at +25 °C up to 95 %; atmospheric pressure 84 to 106.7 kPa.

4 Safety Instructions

Observe general high-voltage workplace safety rules when operating the instrument.

All personnel operating or servicing the instrument must be familiar, to the appropriate extent, with the Russian Federation's electrical installation operating rules (PTEEP, Order of the Ministry of Energy No. 6 of 13.01.2003) and occupational safety rules for electrical installations (POTEU, Order of the Ministry of Labor No. 903N of 15.12.2020), or the equivalent local regulations in the country of use.

DANGER

Grounding Required

Operating the instrument on an ungrounded high-voltage source is prohibited. The grounding terminal must be clearly marked; grounding contact surfaces must be undamaged, clean, smooth and free of oxidation or corrosion.

The instrument casing, controls, measuring leads and accessories must be free of mechanical damage. All connections must be securely made and undamaged; the mains plug's grounding contacts must be in good condition and provide reliable electrical contact.

Only qualified electrical personnel — familiar with this documentation, trained in safe working methods for this equipment, holding electrical safety qualification group III or higher, and authorized for independent work in installations above 1000 V — may operate the instrument. Personnel who have not passed the required assessment must not be permitted to work with the instrument. The workplace must comply with fire-safety requirements (GOST 12.1.004-76 or local equivalent).

5 Scope of Delivery

The instrument is packed in an individual plywood crate. The control unit and high-voltage unit are secured inside the crate with wooden battens to prevent movement. Accessories are packed per the equipment set below, in plastic bags; the documentation is packed in a separate plastic bag.

Table 4 — Scope of delivery

Item	Qty	Remarks
Control unit (per ordered modification)	1	—
High-voltage unit (per ordered modification)	1	—
Interconnecting cable	1	(4±0.1) m
Mains cable	1	(4±0.1) m, 250 V, 16 A
Grounding cable	2	(4±0.1) m, 4 mm ²
Fuse, 15 A	2	Ceramic, 6.35×30 mm
Signal-circuit connector plug	1	FQ14-6TJ
Technical Passport	1	STSK.411728.014-2022 PS
User Manual	1	STSK.411728.014-2022 UM

6 Service Life, Storage Life and Warranty

Service life to the first scheduled maintenance (oil change) is two years; subsequent scheduled maintenance is performed once a year.

Total service life is ten years, inclusive of a storage/preservation period of up to two years (in the manufacturer's original packaging) in warehouse conditions.

These service-life and storage-life figures are valid provided the user complies with the requirements of the current operating documentation.

Warranty period: 18 months from the date of sale.

7 Packaging Certificate

PACKAGING CERTIFICATE	
SKAT Digital Dielectric Test System, STSK.411728.014	
Modification: _____	Serial No.: _____
Packed by IEF SKAT in accordance with the applicable technical documentation.	
Packed by: _____ Position: _____ Signature: _____	

8 Acceptance Certificate

ACCEPTANCE CERTIFICATE	
SKAT Digital Dielectric Test System, STSK.411728.014	
Modification: _____	Serial No.: _____
Manufactured and accepted in accordance with technical specifications STSK.411728.014 TU, and found fit for service.	
QC Manager: _____ Signature: _____ Date: _____	

9 Commissioning Certificate

COMMISSIONING CERTIFICATE	
SKAT Digital Dielectric Test System — placed into service	
Modification: _____	Serial No.: _____
Commissioning date: _____	
Commissioned by: _____ Signature: _____	
Accepted by: _____ Signature: _____	

10 Calibration Record

Calibration entries are logged in Table 5.

Table 5 — Calibration record

Calibration Date	Certificate No.	Result	Technician (name)	Signature & Stamp

11 Maintenance and Repair Log

Maintenance is performed in accordance with User Manual STSK.411728.014-2022 UM. Maintenance and repair events are logged in Table 6.

Table 6 — Maintenance and repair record

Date	Type of Service	Basis (document No. & date)	Performed by	Verified by	Remarks

12 Storage and Transportation Rules

Store the instrument, in its factory packaging, per storage condition 2 of GOST 15150-69 (Table 13), at an ambient temperature from $-50\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$ and relative humidity up to 98 % at $25\text{ }^{\circ}\text{C}$.

Prior to commissioning, store the unpackaged instrument in a clean, dry room at an ambient temperature from $+1\text{ }^{\circ}\text{C}$ to $+35\text{ }^{\circ}\text{C}$ and relative humidity up to 80 % at $25\text{ }^{\circ}\text{C}$.

Transport the packaged instrument by any mode of transport that ensures protection from damage, in accordance with the carriage rules applicable to that mode. Transport conditions with respect to climatic factors must comply with GOST 15150, and the instrument must be carried only in enclosed vehicles, in its factory packaging, at an ambient temperature from $-50\text{ }^{\circ}\text{C}$ to $+50\text{ }^{\circ}\text{C}$ and relative humidity up to 100 % at $25\text{ }^{\circ}\text{C}$. When transported by air, the instrument must be carried only in heated, pressurized compartments.

Secure the packaged instrument on the transport vehicle so that it cannot shift in transit.

13 Disposal

When the instrument reaches the end of its service life and repair is no longer practicable or economically justified, it must be decommissioned and disposed of as follows:

- Drain the transformer oil from the high-voltage transformer's tank into a dedicated container and dispose of it in accordance with the applicable requirements for used oil and lubricants (GOST R 52108-2003 or the equivalent local standard).
- Disassemble the instrument into its component parts and dispose of them per GOST R 52108-2003 or the equivalent local standard.

Revision Record

Rev.	Pages				Document No.	Reference & Date	Signature	Date
	Changed	Superseded	New	Removed				